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Polyethylene Glycol and Polyethylene Glycol/hydroxyapatite Constructs Produced through Stereo-thermal-lithography

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Abstract. Stereo-thermal-lithography (STLG) is an innovative system that uses ultraviolet (UV) radiation and near-infrared (IR) radiation simultaneously to initiate the curing reaction in a liquid solution, containing both UV and IR photoinitiator. In this research work, poly(ethylene glycol) dimethacrylate (PEGDMA) and PEGDMA/hydroxyapatite (HA) constructs were produced using the STLG system, operating in the UV mode, and characterised regarding the morphology and water absorption properties. Constructs were produced with different geometries and shapes by introducing several variations in the processing parameters, such as the irradiation time and the layer thickness. Results show the ability of the system to produce constructs at the micro-scale with very good definition and resolution. The irradiation time is the critical processing parameter, strongly affecting the water absorption and the structural integrity of the constructs.

Introduction

Stereolithography is an additive manufacturing technique with highest accuracy, precision and resolution. It produces 3D solid objects through the irradiation of a liquid photosensitive material, using two distinct methods, the mask-based method and the direct or laser writing method [1-4]. In the mask-based method, an image is transferred to a liquid polymer by irradiating through a patterned mask, containing transparent zones that correspond to the sections of the model to be built. This method enables the curing of an entire layer in just one irradiation step, reducing the fabrication time, and avoiding undesirable polymerizations. The laser writing method involves the use of a focused laser beam to selectively irradiate and solidify the liquid photopolymer [3, 5]. Stereolithographic processes are widely used in the biomedical field, manly to produce 3D scaffolds for tissue regeneration [6, 7], and complex constructs containing multiple materials and cell types, to investigate the cell-cell and the cell-biomaterial interactions [8, 9]. Besides the increasing interest in stereolithographic processes for the fabrication of constructs for biomedical applications, their main limitations concern on the low-throughput, due to the high processing time, the fabrication of single material constructs, and on the lack of available photocurable, biocompatible and biodegradable materials [10].

The STLG system (Fig. 1) intends to solve some of the major limitations of conventional stereolithography, such as efficiency, speed and accuracy [11, 12]. The system combines UV radiation and near-IR radiation, operating in three different modes: subsystem A uses UV radiation to solidify a photopolymer containing a certain amount of UV initiator; the subsystem B uses IR radiation to solidify a photopolymer containing a certain amount of IR initiator; and the subsystem C uses both UV and IR radiation to solidify a photopolymer incorporating a certain amount of both UV and IR initiators. STLG system operates in the mask-based method, using a mercury lamp of 350 W as a light source, and appropriated filters to split the radiation into two different wavelengths: UV radiation and near-IR radiation. Optical fibres, projection and focal lenses irradiate a UV-DMD and a

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IR-DMD. In the case of subsystem C, a dichroic mirror captures the images projected on both DMDs (1024 x 768 pixels, 14 mm in size), combining them into a single image that is transferred to the liquid polymer. The equipment also includes a multi-vat system, enabling the fabrication of multimaterial constructs. The vertical displacement of the platform is secured by a positioner uniaxial MYCOSIS® Translation Stage VT-80, which allows vertical increments of 1 μm, at a speed ranging between 0.001 and 20 mm/s. Its main advantages over other stereolithographic processes are the following ones: (i) a more efficient generation of radicals, due to the combination of multiple light effects; (ii) the use of small concentrations of the two types of initiator (UV and near-IR initiators), enabling the radiation to penetrate deeper into the polymer; and (iii) a more localised curing reaction, improving the accuracy of the produced models. A home-made software enables the control over the optical, cleaning and construction units.

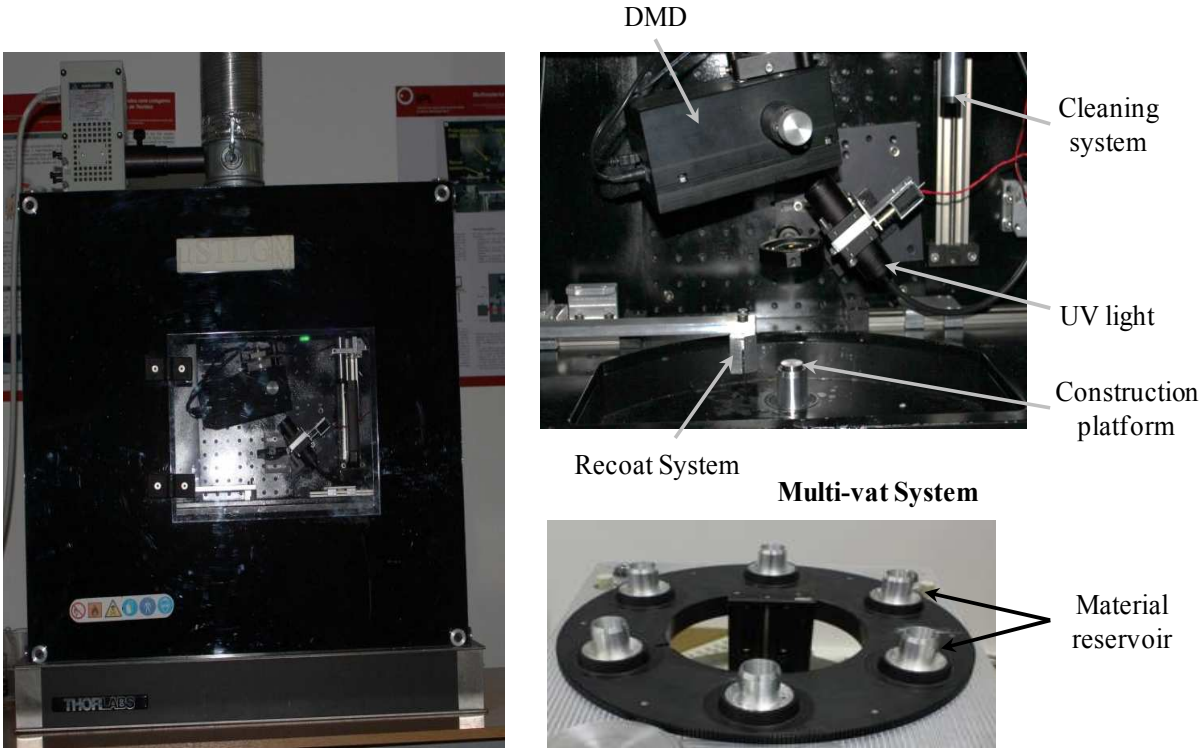


Fig. 1. STLG system and its main optical and mechanical components.

This paper describes the fabrication of polymeric (PEGDMA) and composite (PEGDMA/HA) constructs using the STLG system. The properties of the produced constructs were evaluated through scanning electron microscopy (SEM), energy dispersive spectroscopy (EDS) and water absorption tests.

Experimental Procedure

Photopolymer solution. Poly(ethylene glycol) dimethacrylate (PEGDMA, MW 750; Sigma Aldrich, Portugal) was dissolved in distilled water at a concentration of 60% (v/v), and kept under constant stirring (600rpm), during 2 hours. Then, the photoinitiator 2-Hydroxy-4′-(2-hydroxyethoxy)-2-methylpropiophenone (Irgacure 2959, Sigma Aldrich, Portugal) was added to the photopolymer solution (10%, w/v), and kept under stirring (400rpm), during 12 hours. In order to prepare composite constructs, the bioactive ceramic hydroxyapatite (HA, <6μm particle size, Altakitin, Portugal) was added to the photopolymer solution at a concentration of 2.5% (w/v), during preparation.

Fabrication of 3D constructs. The constructs were produced using the STLG system operating in the UV mode, as previously described [11, 12]. The photopolymer solutions were irradiated using a UV light (λmax = 365nm and a light intensity of 50mV/cm2) and the parameters indicated in Table 1.

Layers were produced by UV mask irradiation through a UV-DMD system (DMD Discovery™ 1100 Controller Board, Texas Instruments). A home-made software was used to control the fabrication process, enabling the definition of the fabrication parameters such as the irradiation time, the layer thickness and the number layer.

Table 1. Fabrication parameters used to produce the constructs.

Photopolymer solution	Irradiation Time (IT) (seconds)	Layer Thickness (LT) (μm)	Layer number
PEGDMA	45	125	3
	60		
	90		
	60	62.5	
		250	
		500	

Morphological characterisation. The surface morphology was investigated through the SEM analysis, using the Hitachi S-2700 SEM system (Scanservice Corporation, California) at 20kV. The dimensions of the constructs were measured using the software Image J.

Energy Dispersive Spectroscopy (EDS) analysis. The EDS analysis was performed to evaluate the presence of calcium (Ca) and phosphorus (P) deposits in the surface of both PEGDMA and PEGDMA/HA constructs. The tests were carried out in the EDS system (Rontec USA, Inc.).

Water absorption capacity. To investigate the effect of the irradiation time (15, 25, 35, 45 and 60s) on the water absorption properties, PEGDMA constructs with a diameter of 0.5 cm were produced, and dried at 25ºC during 48 hours to determine the dry weight. Then, the samples were immersed in distilled water at physiologic temperature (37ºC), during 24 hours. Finally, the samples were removed from the solution, the excess of water at surface removed with a filter paper, and the hydrated weight immediately accessed. The normalized water absorption was determined using the Eq. 1:

$$\text{Water absorption} = W_h / W_d \quad (1)$$

where, Wh and Wd correspond to the weight of the samples before and after immersion into distilled water, respectively. Four specimens were used for each condition.

Results and Discussion

Morphological characterisation. Fig. 2 shows the influence of the irradiation time on the surface morphology of the PEGDMA constructs. SEM micrographs indicate that all constructs exhibit a well-defined geometry without non-polymerized regions in the centre of the structures. In the construct irradiated during 45s, it is possible to observe the presence of non-polymerized regions, due to the insufficient curing time. As expected, the increase in the irradiation time produces constructs without significant defects and non-polymerized regions.

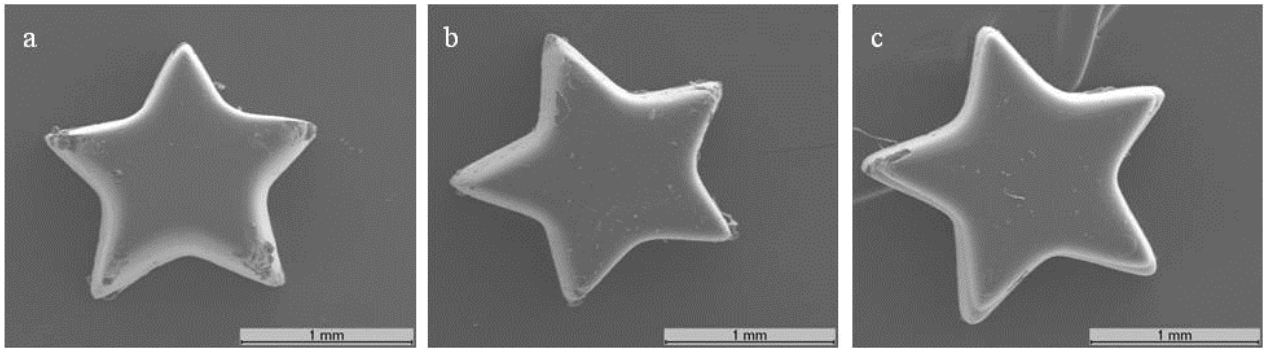


Fig. 2. SEM micrographs of PEGDMA constructs produced by STLG system, using different irradiation times: IR= 45s (a), IR = 60s (b), IR = 90s (c).

Fig. 3 presents the SEM micrographs of the PEGDMA constructs produced using different layer thickness values. Results show that all constructs present a well-defined geometry without significant defects. From the SEM images, it is also possible to observe that the constructs exhibit a good adhesion between layers without delamination between adjacent layers.

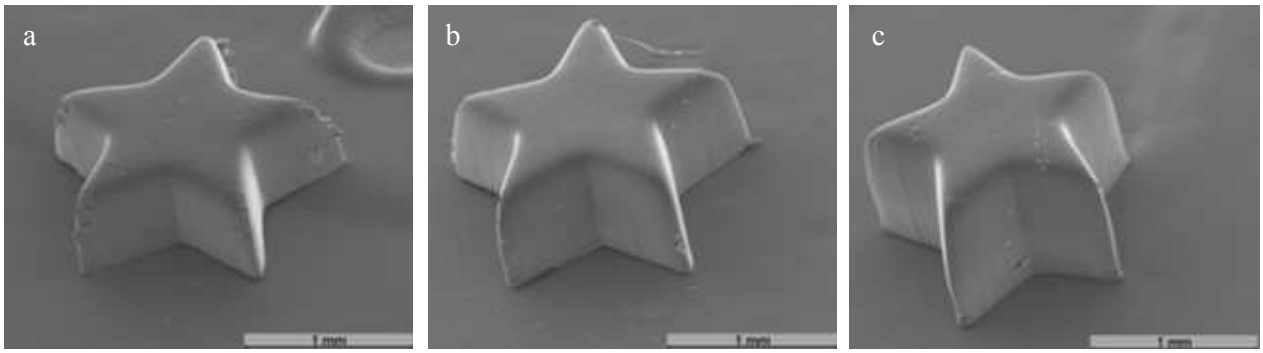


Fig. 3. SEM micrographs of PEGDMA constructs produced using different layer thicknesses: LT = 62.5µm (a), sample height = 488 ± 2.31µm; LT = 250µm (b), sample height = 573 ± 52.53µm; LT = 500µm, sample height = 676 ± 4.36µm (c).

In order to test the ability of the STLG system to produce 3D constructs for tissue engineering, scaffolds with different architectures were produced using the PEGDMA photopolymer solution. Scaffolds with a lay down pattern of 0º/90º and 0º/45º/90º, were fabricated using an irradiation time of 60s, and a layer thickness of 125µm. The PEGDMA scaffolds exhibit a uniform surface morphology, a well-defined internal geometry without polymerized regions within the pores, and an uniform pore distribution (Fig. 4). SEM images also indicate that the scaffolds present uniform filaments with an average thickness of 94.67 ± 4.62 µm (0º/90º) and 97.67 ± 7.51µm (0º/45º/90º).

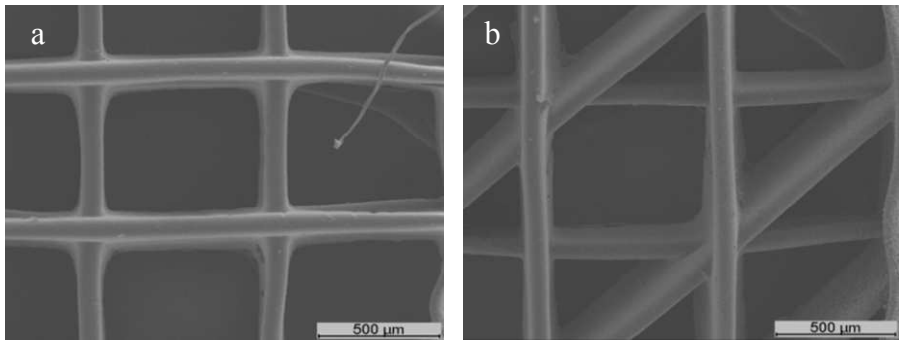


Fig. 4. PEGDMA hydrogel scaffolds with a lay down pattern of 0º/90º (a) and 0º/45º/90º (b).

Energy Dispersive Spectroscopy (EDS) analysis. Energy dispersive spectroscopy was used to quantify the elemental composition of HA into the PEGDMA and PEGDMA/HA scaffolds (Fig. 5). Results show that the main elements present in the surface of the PEGDMA scaffolds include the carbon (C) and the oxygen (O), which are present in the chemical composition of the photopolymer. On the other hand, the PEGDMA/HA hydrogel scaffolds present four characteristic peaks, corresponding to the C, O, P, and Ca elements. The presence of the P and Ca elements in the composite scaffold confirms the present of the HA, because these elements are part of its chemical composition ([Ca5(OH)(PO4)3]x). EDS results also show the absence of impurities like carbonaceous species within the HA nanoparticles. The SEM micrographs, indicated in Fig. 5, show that the composite scaffolds exhibit a rougher surface than the PEGDMA scaffolds, due to the presence of the HA nanoparticles at the surface of the filaments. These results indicate that the STLG system enables the fabrication of 3D scaffolds containing bioactive elements, which is of particular interest for tissue engineering applications.

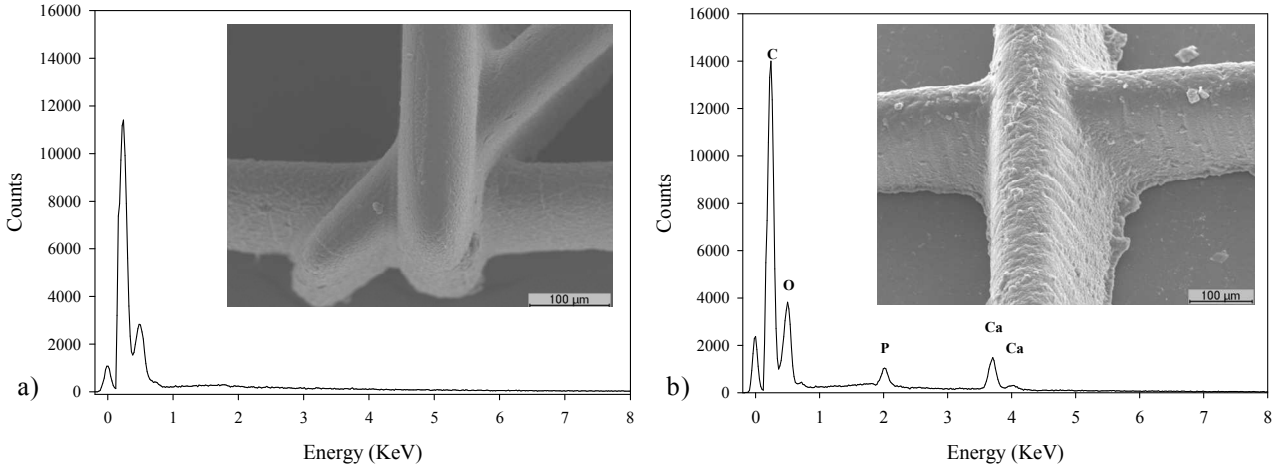


Fig. 5. EDS analysis and SEM micrographs of the PEGDMA (a) and PEGDMA/HA hydrogel scaffolds (b).

Water absorption capacity. Fig. 6 shows the water absorption capacity of PEGDMA constructs, using different irradiation times. The use of low polymerization times, namely 15s and 25s, produce a fragile structures, which contain a high sol fraction and are difficult to handle, due to the incomplete curing reaction. On the other hand, the use of higher irradiation times (35s, 45s and 60s) result on constructs with good structural integrity, easy to handle and showing well-defined geometry. Regarding the water absorption test, results show that the constructs irradiated during 15s and 25s seem to present reduced values of water absorption, due to the construct dissolution (non-completed curing reaction), which reduces the available area to absorb and retain water. The increase in the irradiation time allows to obtain constructs with a more stable 3D network, exhibiting water absorption values in the range of 1.70-1.76, without significant differences between them. After the water absorption test, samples irradiated during 15s, exhibit a fragile structure with polymer loss as a result of the dissolution and fragmentation phenomena. Contrary, the samples irradiated during 60s, maintain the initial shape without significant changes in both structural integrity and weight loss, as shown in Fig. 6.

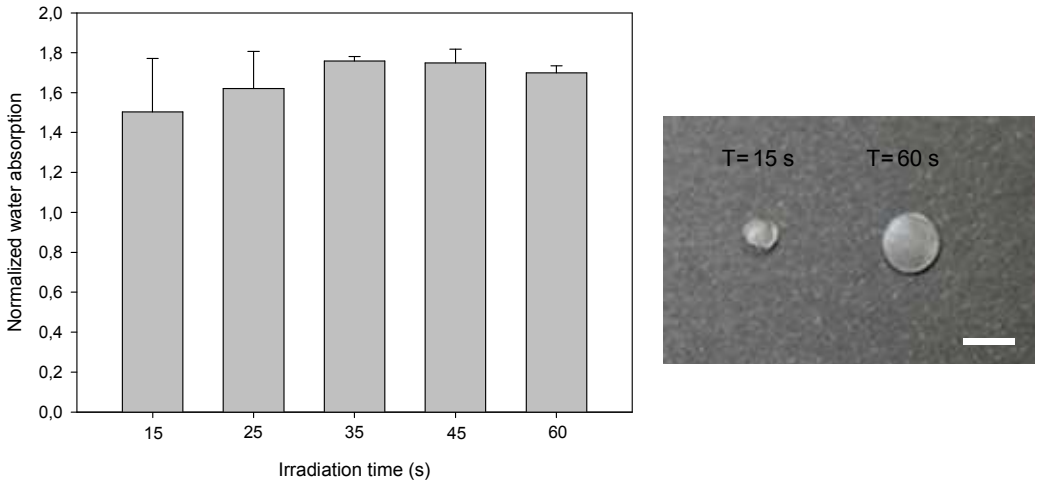


Fig. 6. Influence of the irradiation time on the water absorption of PEGDMA constructs built by STLG system (left). Macroscopic images of the constructs produced using 15s and 60s of irradiation time, after immersion into distilled water during 24 hours (right). Scale bar represents 0.5mm.

Conclusions

The STLG system enables the fabrication of hydrogel constructs with a well-defined geometry and resolution. Results show that the irradiation time strongly affects both the surface morphology and structural integrity of the hydrogel constructs. EDS analysis indicates the presence of the bioactive compounds of HA on the surface of PEGDMA/HA hydrogel scaffolds.

Acknowledgements

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